Elementary row operations and elementary column operations on a matrix are known as elementary transformations.

We use the following notations for elementary row transformations:

(i) Interchanging of $i^{th}$ and $j^{th}$ rows is denoted by $R_i \leftrightarrow R_j$.

(ii) The multiplication of each element of $i^{th}$ row by a non-zero constant $\lambda$ is denoted by $R_i \to \lambda R_i$.

(iii) Addition to $i^{th}$ row, a non-zero constant $\lambda$ multiple of $j^{th}$ row is denoted by $R_i \to R_i + \lambda R_j$.

Similar notations are used for elementary column transformations.

**Definition 1.4**

Two matrices $A$ and $B$ of same order are said to be equivalent to one another if one can be obtained from the other by the applications of elementary transformations. Symbolically, we write $A \sim B$ to mean that the matrix $A$ is equivalent to the matrix $B$.

For instance, let us consider a matrix $A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$.

After performing the elementary row operation $R_2 \to R_2 + R_1$ on $A$, we get a matrix $B$ in which the second row is the sum of the second row in $A$ and the first row in $A$.

Thus, we get $A \sim B = \begin{bmatrix} 1 & -2 & 2 \\ 0 & -1 & 5 \\ 1 & -1 & -4 \end{bmatrix}$.

The above elementary row transformation is also represented as follows:

$$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} \xrightarrow{R_2 \to R_2 + R_1} \begin{bmatrix} 1 & -2 & 2 \\ 0 & -1 & 5 \\ 1 & -1 & -4 \end{bmatrix}.$$

**Note**

An elementary transformation transforms a given matrix into another matrix which need not be equal to the given matrix.

## 1.3.2 Row-Echelon form

Using the row elementary operations, we can transform a given non-zero matrix to a simplified form called a **Row-echelon form**. In a row-echelon form, we may have rows all of whose entries are zero. Such rows are called **zero rows**. A non-zero row is one in which at least one of the entries is not zero. For instance, in the matrix $\begin{bmatrix} 6 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $R_1$ and $R_2$ are non-zero rows and $R_3$ is a zero row.

**Definition 1.5**

A non-zero matrix $E$ is said to be in a **row-echelon form** if:

(i) All zero rows of $E$ occur below every non-zero row of $E$.

(ii) The first non-zero element in any row $i$ of $E$ occurs in the $j^{th}$ column of $E$, then all other entries in the $j^{th}$ column of $E$ below the first non-zero element of row $i$ are zeros.

(iii) The first non-zero entry in the $i^{th}$ row of $E$ lies to the left of the first non-zero entry in $(i+1)^{th}$ row of $E$.



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